

Probabilistic Forecasting and Trading on Event Markets:

A Case Study of Daily Temperature Contracts at San Francisco International Airport

A PREPRINT

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May 4, 2026

Abstract

*Event-contract exchanges such as Kalshi have, since their CFTC designation as designated contract markets, transformed several previously informal categories of uncertainty—including daily temperature, snowfall, and rainfall at major U.S. airports—into liquid binary financial instruments. We treat the resulting contract chains as discretizations of a conditional probability distribution and ask whether a station-specific statistical learner can systematically price gaps between its own distributional forecasts and market-implied probabilities. We construct a fully reproducible system for the San Francisco International Airport (SFO) daily-temperature markets using only publicly available data: **56** years of NOAA Local Climatological Data and Integrated Surface Database observations (1970–2026, 493,569 hourly rows after merging), the public Kalshi market-data API (612 settled markets, 24,502 hourly candles), and the National Weather Service Aviation Weather METAR feed for live updates. Hourly temperatures are forecast at seven horizons by a multi-quantile gradient-boosting model (49 models in total) with native missing-value handling and station-specific marine-layer features; daily-extreme contracts are addressed by a separate, target-specific quantile model that conditions on hours-to-settlement. The predicted distribution is mapped onto each Kalshi strike via cumulative-distribution interpolation that respects the exchange’s integer-degree settlement convention, producing a per-bucket probability that is then refined by a small logistic meta-calibrator blended with the market’s mid price.*

*On the held-out test set (28,977 hours, 2023–2026), our hourly point forecasts attain mean absolute errors ranging from **0.86** °F at +1h to **3.00** °F at +72h, with consistently near-zero bias and skill scores of +30% to +75% against persistence and +13% to +75% against climatology. On Kalshi data ($n = 11,953$ decision-time observations), the model’s probabilistic skill is sharply horizon-dependent: at ≤ 6 hours to settlement we attain log-loss 0.118 versus the market’s 0.812 (a +85.5% skill score), while at 24–48 hours the market dominates (−84.8%). A realistic trading backtest with quadratic fees, spread crossing, position caps and fractional Kelly sizing (maximum stake \$200/trade,*

$\leq 5\%$ bankroll) shows a horizon-filtered strategy ($\leq 12h$ to settle) producing a +723% return on a \$1,000 initial bankroll over 108 days with 277 trades and a 65% win rate; restricting further to $\leq 6h$ yields 78% win rate at \$41 profit per trade. We argue these results characterize event markets as a distinct asset class whose pricing inefficiencies sit at the join of forecast post-processing and market microstructure rather than at either alone, and we close by outlining the limits of the strategy and its natural extensions.

Keywords: prediction markets, probabilistic forecasting, quantile regression, gradient boosting, weather contracts, market microstructure, event markets, calibration

1 Introduction

The premise of an event-contract exchange is straightforward. A future event is specified precisely enough to be settled on a public, auditable record (a National Weather Service climatological report; the score of a basketball game; an official election certification), the exchange lists a chain of mutually-exclusive binary contracts that partition the outcome space, and participants trade those contracts continuously until settlement. The CFTC’s designation of Kalshi as a designated contract market in 2020, and the subsequent expansion of contract categories from political and macroeconomic events to weather, sports, and entertainment, has accelerated the growth of this market structure considerably.

We focus in this paper on a single, narrow vertical: the daily high and daily low temperature contracts settled on the San Francisco International Airport (SFO) station. There are several reasons to begin here. First, the underlying truth source is unambiguous: the National Weather Service Climatological Report for SFO publishes the official daily extremes, and these are the explicit settlement source of the relevant Kalshi contracts. Second, the same agency publishes more than a half-century of hourly observations for the same station through two complementary archives: the Local Climatological Data product [National Centers for Environmental Information, 2025b] and the Integrated Surface Database [National Centers for Environmental Information, 2025a]. Third, multiple decades of literature on probabilistic weather forecasting [Gneiting and Katzfuss, 2014, Rasp and Lerch, 2018, Taillardat et al., 2016, Brier, 1950] provide a clear methodological foundation: the forecast object is a predictive distribution, not a point estimate, and its quality is measured by proper scoring rules [Gneiting and Raftery, 2007]. Fourth, by working at a single station with strong, well-understood diurnal and seasonal structure, we can isolate the effects of *forecast-market disagreement* from those of forecasting skill itself.

The empirical story we tell in the remainder of this paper has three parts.

A reproducible probabilistic forecasting pipeline. Section 4 documents how we merge the LCD and ISD archives, including a previously unreported timezone-alignment issue between the two products that cleanly explains a class of low-frequency residuals in naive merges. Section 5 describes the model: gradient-boosted multi-quantile regression at fixed forecast horizons, a separate daily-extreme model conditioning on hours-to-settlement, and a 250-feature engineering pipeline including station-specific marine-layer proxies. Held-out validation indicates well-calibrated quantile predictions across all horizons, and the test set (2023–2026) yields mean absolute errors of 0.86 °F at +1h growing to 3.00 °F at +72h.

A principled mapping from distribution to strike. Section 6 describes how we transform a predicted hourly or daily-extreme distribution into per-strike probabilities for each Kalshi contract chain. Three subtleties surface: (i) the exchange’s integer-rounding convention requires bucket probabilities to be computed as differences of the predictive CDF at half-degree edges; (ii) raw quantile predictions can cross, requiring monotonization before CDF interpolation; (iii) the model’s edge versus the market is heterogeneous in both bucket and horizon, motivating a small meta-calibrator that blends model and market probabilities via a logistic regression with isotonic post-correction.

An honest trading backtest. Section 7 simulates trading under realistic execution constraints—quadratic Kalshi fees, mandatory spread-crossing, position caps proportional to candle-time open-interest, fractional-Kelly sizing capped at 5% of current bankroll, and a slippage model proportional to size beyond top-of-book. All horizons on a \$1,000 initial bankroll yields a moderate negative return (−37%) because the market’s forecasting advantage dominates at long horizons. Restricting to decisions within 12 hours of settlement reverses this: +723% over 108 days with a 65% win rate. Restricting further to ≤ 6 hours raises the win rate to 78% at \$41 profit per trade but trades fewer markets.

We organize the discussion (Section 9) around what these findings suggest about event markets more broadly. The combination of short-horizon forecasting skill and rapid market convergence implies that the exploitable inefficiency in this asset class is not, primarily, the ability to forecast better than professional meteorologists; it is the ability to integrate the most recent observations into prices faster than the market does. We believe this characterization—small, asymmetric, observation-driven edges on top of a broadly accurate market—generalizes to other event-contract verticals where the truth source is similarly observable shortly before settlement.

The full software pipeline, all parquet outputs, and all 63 trained models (49 hourly + 14 daily-extreme) are reproducible from the public NOAA and Kalshi APIs and the shell of code accompanying this paper.

2 A systems perspective on event markets

It is useful, before turning to specific data and models, to fix a high-level view of what an event-contract exchange is, mathematically, and what would constitute exploitable inefficiency in such an exchange.

2.1 Event markets as a feedback loop

A liquid event market couples five layers (*event reality*, *data streams*, *forecast models*, *market prices*, *capital allocation*) into a recursive feedback loop. Reality generates data; data trains forecasting models; models inform prices; prices clear capital allocation; capital allocation drives the construction of additional data infrastructure that improves the next round of forecasting.

$$\underbrace{Y_E \in \{0, 1\}}_{\text{event reality}} \longrightarrow \underbrace{\mathcal{I}_t}_{\text{data streams}} \longrightarrow \underbrace{p_\theta(\cdot | \mathcal{I}_t)}_{\text{forecast model}} \longrightarrow \underbrace{c_t}_{\text{market price}} \longrightarrow \underbrace{\Pi_t = c_t \cdot \mathbb{1}_{Y_E}}_{\text{capital flow}}.$$

The market-clearing layer can be approximated by the assumption that the price c_t implements

$$c_t = \mathbb{E}[Y_E \mid \mathcal{F}_t], \quad (1)$$

where $\mathcal{F}_t$ is the public information set at time t . Equation (1) is exact under the no-arbitrage assumption of a complete frictionless market with a risk-neutral participant [Wolfer and Zitzewitz, 2004]; it is approximate in real exchanges because of fees, transaction costs, risk-averse participants, and incomplete information aggregation [Manski, 2006].

The exploitable inefficiency in any specific market is the gap

$$\Delta_t = p_\theta(Y_E = 1 \mid \mathcal{I}_t^\theta) - c_t, \quad (2)$$

where $\mathcal{I}_t^\theta$ is the modeler’s information set (potentially a strict superset, subset, or intersection of $\mathcal{F}_t$). When $\mathcal{I}_t^\theta \subset \mathcal{F}_t$ —the modeler has *less* information than the market—we expect $\Delta_t \rightarrow 0$ in mean and $\mathbb{E}[\Delta_t \cdot (Y_E - c_t)] \rightarrow 0$, reflecting the fact that the modeler cannot systematically improve on a strict-superset price. When $\mathcal{I}_t^\theta \supset \mathcal{F}_t$ —the modeler has access to information the market does not—arbitrage in expectation is positive. The intermediate case, with neither inclusion strict, is the operationally common one and the one we analyze empirically.

2.2 The asset-class transition

A frequently-observed pattern is that informal categories of uncertainty become liquid asset classes through five stages: *specification* (contracts standardized), *data* (outcome and feature streams documented), *liquidity* (continuous market depth), *institutional participation* (market-making, hedging, derivatives), and *risk transfer* (use by entities with real economic exposure to the underlying). The pattern is well-documented in classical financial markets [Maccheroni et al., 2009, López de Prado, 2018] and in the older weather-derivatives literature [Hong et al., 2016].

Daily-temperature event contracts on Kalshi are at a distinctive point in this transition. Specification (stage 1) and data (stage 2) are mature: contract terms are publicly documented and the underlying NOAA archives exceed 50 years. Liquidity (stage 3) is partially developed—the markets trade actively in the days surrounding settlement but with notable microstructural inefficiencies in price convergence. Institutional participation (stage 4) is limited; reports of institutional liquidity provision in the broader event-market space exist, but the magnitude is small relative to traditional markets. Risk transfer (stage 5) is largely absent for individual contract chains, although weather-derivative use by energy companies and agricultural producers [National Centers for Environmental Information, 2025b] is well-documented for adjacent products.

The strategy described in this paper exploits stage-3 inefficiencies specifically: a market that has standardized contracts and good data but has not yet developed the depth or participant mix to integrate short-horizon observations with the speed seen in mature financial markets.

3 Background and related work

3.1 Probabilistic weather forecasting and post-processing

The modern theory of probabilistic forecasting is built around proper scoring rules: scoring functions whose expected value is minimized at the true predictive distribution [Gneiting and Raftery, 2007,

Gneiting and Katzfuss, 2014, Wilks, 2019]. The pinball (quantile) loss

$$\text{PL}_\tau(y, \hat{q}) = \begin{cases} \tau(y - \hat{q}), & y \geq \hat{q} \\ (\tau - 1)(y - \hat{q}), & y < \hat{q} \end{cases} = (\tau - \mathbb{I}[y < \hat{q}]) \cdot (y - \hat{q}) \quad (3)$$

is consistent for the τ -quantile of the conditional distribution [Koenker and Bassett Jr, 1978]; minimizing $\mathbb{E}[\text{PL}_\tau]$ recovers exactly $F_{Y|X}^{-1}(\tau)$. Aggregated over a range of quantiles, this score approximates the Continuous Ranked Probability Score (CRPS) [Matheson and Winkler, 1976, Hersbach, 2000], defined for a predictive CDF $\hat{F}$ and observation y by

$$\text{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} (\hat{F}(x) - \mathbb{I}[y \leq x])^2 dx = \int_0^1 \text{PL}_\tau(y, \hat{F}^{-1}(\tau)) d\tau, \quad (4)$$

the second equality being the well-known equivalence between CRPS and the integral of the pinball loss [Gneiting and Katzfuss, 2014]. The Brier score [Brier, 1950] for a binary outcome $Y \in \{0, 1\}$ and forecast probability p is

$$\text{Br}(p, y) = (p - y)^2, \quad (5)$$

and admits the Murphy decomposition [Murphy, 1973] into reliability, resolution, and uncertainty components,

$$\mathbb{E}[\text{Br}] = \underbrace{\mathbb{E}[(p - \mathbb{E}[Y | p])^2]}_{\text{reliability (calibration)}} - \underbrace{\mathbb{E}[(\mathbb{E}[Y | p] - \bar{Y})^2]}_{\text{resolution}} + \underbrace{\bar{Y}(1 - \bar{Y})}_{\text{uncertainty}}.$$

We use the pinball loss for training, the Brier score and log-loss for evaluation.

For weather forecasts specifically, post-processing of numerical weather prediction (NWP) output is a mature subfield. Quantile-regression-forest methods [Taillardat et al., 2016] and neural-network ensemble model output statistics [Rasp and Lerch, 2018] have shown consistent improvements over raw ensemble forecasts for surface variables. Recent end-to-end neural weather models such as Pangu-Weather [Bi et al., 2023], FourCastNet [Pathak et al., 2022], and GraphCast [Lam et al., 2023] have demonstrated that data-driven medium-range forecasts can match or exceed operational NWP models on standard verification metrics. Our contribution is orthogonal: we do not develop a new global forecasting model, but rather focus on a single station, the relationship between its forecast distribution and an event-market price, and the trading strategy that arises.

3.2 Quantile gradient boosting

Gradient boosting [Friedman, 2001, Breiman, 2001] extended naturally from squared-error loss to quantile loss and has been a workhorse for tabular probabilistic forecasting in fields ranging from energy demand [Hong et al., 2016] to climate-derivative pricing. The functional formulation is

$$\hat{q}_\tau^{(T)}(x) = \sum_{m=1}^T \nu \cdot h_m(x), \quad h_m = \arg \min_{h \in \mathcal{H}} \sum_i \text{PL}_\tau(y_i, \hat{q}_\tau^{(m-1)}(x_i) + h(x_i)), \quad (6)$$

where h_m is a regression tree with bounded leaf count $|\mathcal{H}|$ and $\nu \in (0, 1]$ is the learning rate. Modern implementations including LightGBM [Ke et al., 2017], XGBoost [Chen and Guestrin, 2016], and CatBoost [Prokhorenkova et al., 2018] provide quantile-loss training, native missing-value handling, and histogram-based tree induction, all of which are useful in the present setting. We use scikit-learn’s `HistGradientBoostingRegressor` [Pedregosa et al., 2011], which implements histogram-based gradient boosting with native NaN handling, both because it removes the need for imputation (the fact of missingness is itself signal in our setting) and because it requires no system-level dependencies beyond a Python interpreter.

3.3 Prediction markets and information aggregation

Prediction markets have a long theoretical lineage, from Hayek’s argument that prices aggregate dispersed knowledge [Hayek, 1945] to Plott and Sunder’s experimental verification of rational-expectations information aggregation in laboratory markets [Plott and Sunder, 1988], through Hanson’s logarithmic market-scoring rule (LMSR) which provides an explicit cost function for an automated market maker [Hanson, 2003, 2007]. The LMSR’s cost function for an event with K mutually exclusive outcomes and outstanding share vector $\mathbf{q} = (q_1, \dots, q_K)$ is

$$C(\mathbf{q}) = b \log \sum_{k=1}^K \exp(q_k/b), \quad p_k(\mathbf{q}) = \frac{\partial C}{\partial q_k} = \frac{e^{q_k/b}}{\sum_j e^{q_j/b}}, \quad (7)$$

parameterized by a liquidity scale $b > 0$. The implied probabilities $\{p_k\}$ form a softmax of the share vector and are, by construction, the conditional expectations of the indicator variables $\mathbb{I}[Y_E = k]$ under Hanson’s market-maker. Equation (7) is the formal foundation of the equivalence we use throughout this paper between event-contract prices and conditional probabilities.

The empirical literature is large. Forsythe et al. [1992] and Berg et al. [2008] document the predictive accuracy of the Iowa Electronic Markets across a dozen years of presidential elections; Snowberg et al. [2007] and Leigh and Wolfers [2013] extend the analysis to economic policy and foreign-exchange markets respectively; Ali [1977], Ottaviani and Sørensen [2009] provide foundational work on betting-market efficiency. Wolfers and Zitzewitz [2004] provides a now-canonical review. Manski [2006] distinguishes between price and probability in a market with heterogeneous, risk-averse traders—a distinction we will return to when we observe over-rounds (sums of YES prices across mutually-exclusive contracts exceeding one) in Kalshi prices late in a contract’s life. Taylor and Yu [2017] examines price discovery in weather-derivative markets specifically. Commodity Futures Trading Commission [2024] documents the U.S. CFTC regulatory framework that designates Kalshi a contract market.

The empirical literature on event-market efficiency has historically been limited by data: prior to liquid centralized exchanges, prediction-market data was either thin (Iowa Electronic Markets) or scattered across betting exchanges. The growth of Kalshi and similar platforms since 2020 changes this picture; with public market-data APIs, full settlement records, and hourly candle history [Kalshi, 2026], even a single-machine researcher can construct a complete contract-by-contract panel.

3.4 Calibration: post-hoc methods and proper diagnostics

A predictor is well-calibrated if among events to which it assigns probability p , the empirical relative frequency of occurrence is also approximately p . Calibration is necessary for the per-bucket probabilities to be useful for trading: a model that systematically assigns 0.7 to events that occur 0.4 of the time will lose money even if its rank ordering is correct. The classical post-hoc methods are Platt scaling [Platt, 1999] (a sigmoidal fit, useful for support-vector machines and similar margin-based predictors) and isotonic regression [Zadrozny and Elkan, 2002], with the latter usually preferred for forecasters whose miscalibration is non-monotone. Niculescu-Mizil and Caruana [2005] provides systematic empirical comparisons; for tree-based predictors of the sort considered here, isotonic regression usually dominates Platt at sufficient training-set size.

A separate strand of work—conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2021]—

constructs prediction intervals with finite-sample distribution-free coverage guarantees by calibrating a non-conformity score on a held-out exchangeable set. We do not use conformal prediction in the present paper because exchangeability fails for time-series data without explicit blocking, but we note that recent work on online conformal prediction is directly applicable to our setting and is a natural extension.

3.5 Event markets as an emerging asset class

A second relevant literature studies the institutionalization of new asset classes. The general pattern is well-documented: an informal risk—weather, credit, volatility, freight—becomes an asset class when its outcomes are standardized into transferable contracts, when liquid markets form to absorb risk transfer, when historical and real-time data are widely available, and when institutional participants provide market-making and hedging services [Maccheroni et al., 2009, López de Prado, 2018]. Weather derivatives, the closest pre-existing analogue to our contracts of interest, illustrate the pattern: the Chicago Mercantile Exchange has listed heating-degree-day and cooling-degree-day futures and options on a fixed set of stations since 1999, with a vendor ecosystem providing forecasting and pricing services around them [Hong et al., 2016]. The Actuaries Climate Index and similar composite indicators [Taillardat et al., 2016] have likewise taken weather risk and packaged it for actuarial and financial use.

Event-contract exchanges such as Kalshi extend this pattern in two ways. First, they radically lower the granularity of contract specification: rather than monthly heating-degree-day futures, the exchange lists daily high-temperature contracts in 2°F buckets at named airport stations. Second, they shift the participant mix toward retail. Both changes are favorable to the kind of statistical-microstructure strategy we describe: high-granularity contracts mean that even modest station-specific information has a market in which to trade, and a retail-heavy participant mix is the population most likely to leave short-horizon information-incorporation lags unarbitraged.

The absence of NWP-forecast features in our model is therefore both a limitation (long-horizon edge is conceded to the market) and a feature (short-horizon edge is preserved precisely because we do not recapitulate information already in the market price). A research-grade forecast that incorporated GFS, HRRR, and ECMWF outputs would compete with the market price’s information content rather than complement it.

3.6 Risk management and Kelly sizing

For a binary contract priced at $c \in (0, 1)$ with true payout probability p , the Kelly criterion [Kelly, 1956, Merton, 1969] prescribes the bankroll fraction

$$f^* = \arg \max_f \mathbb{E}[\log(1 + fX)] = \frac{p \cdot b - (1 - p)}{b}, \quad b = \frac{1 - c}{c}, \quad (8)$$

where X is the per-dollar payout (random variable taking values $+b = (1 - c)/c$ with probability p and -1 with probability $1 - p$). The objective in (8) maximizes the geometric growth rate of the bankroll W_n under repeated independent identically-distributed bets: $\log W_n = \log W_0 + \sum_{i=1}^n \log(1 + fX_i)$ implies $\frac{1}{n} \log(W_n/W_0) \xrightarrow{a.s.} \mathbb{E}[\log(1 + fX)]$ by the strong law of large numbers, which is maximized at $f = f^*$. The generalization to multiple simultaneous bets requires accounting for the correlation structure of payoffs; in our case, the bets within a single event are perfectly

negatively correlated because the buckets are mutually exclusive and exhaustive, so we treat them as a single portfolio decision rather than independent stakes (Section 7). Practical implementation usually uses fractional Kelly to reduce variance and tail risk [Thorp, 2006]. We apply Kelly with multiplier 0.25 and a hard 5%-of-bankroll cap throughout, both because empirical residuals are fat-tailed in ways that violate Kelly’s stated assumptions [Kahneman and Tversky, 1979] and because real exchange liquidity bounds aggressive sizing regardless of theoretical prescription.

4 Data

4.1 Source archives

Two complementary NOAA archives provide the long-history backbone of our dataset, both for the SFO station (call sign KSFO, ICAO KSFO, NCEI station identifiers USW00023234 for LCD and 72494023234 for ISD).

Local Climatological Data v2 (LCD): 675,469 rows from 1970-01-01 to 2026-04-22, providing hourly observations of dry-bulb temperature, dew point, wet-bulb temperature, relative humidity, sea-level pressure, pressure tendency, sky conditions, visibility, wind direction, wind speed, gust speed, and precipitation, plus daily Summary-of-Day rows giving the official daily maximum, minimum, and average dry-bulb temperatures and daily total precipitation [National Centers for Environmental Information, 2025b]. LCD timestamps are reported in *Local Standard Time* for the station, never adjusted for daylight saving (i.e. Pacific Standard Time year-round at SFO).

Integrated Surface Database (ISD): 569,244 rows from 1973-01-01 to 2025-08-27, providing the same station’s hourly temperature field (TMP) in tenths of degrees Celsius along with quality-control flags [National Centers for Environmental Information, 2025a]. ISD timestamps are in UTC.

Kalshi market data: via the public market-data endpoints documented at Kalshi [2026], we enumerated the SFO weather series KXHIGHTSFO (San Francisco daily HIGH temperature) and KXLWTSFO (daily LOW), recovering 143 events (1 per calendar day, with strike-date metadata), 612 markets (6 buckets per event for HIGH and LOW), and 24,502 hourly OHLC + bid-ask + open-interest candles from each market’s open through settlement. The earliest event in our sample is 2026-01-15 (HIGH) and 2026-04-04 (LOW); the latest is 2026-05-05 in both. No authentication is required for these endpoints—the user’s API key is irrelevant to data access.

Live observations: for real-time pipeline operation we additionally fetch the last 168 hours of KSFO METARs from the NWS Aviation Weather Center JSON API [National Weather Service, 2025] and merge them into our hourly grid on every auto-refresh cycle. This closes the freshness gap between the quarterly-updated LCD archive and the trading window.

4.2 The timezone alignment problem

Because LCD timestamps are in PST and ISD timestamps are in UTC, a naive merge that treats both as wall-clock times places approximately 3.16% of `temp_f` values—those filled by ISD where LCD has a gap—into the wrong calendar hour, off by exactly eight hours. The artifact is easy to detect once one knows to look for it: comparing LCD and ISD hourly readings for any specific date shows the LCD value tracking a typical overnight-to-afternoon trajectory and the ISD value tracking a typical afternoon-to-overnight trajectory, with the two correlated at lag eight hours rather than

zero. The pipeline reported here converts ISD timestamps to PST during ingest, eliminating the artifact and aligning all downstream timestamps on a single PST axis. This finding—that the two flagship NOAA archives use different time conventions and silently break naive merges—is, to the best of our knowledge, not stated explicitly in either archive’s documentation and is worth flagging for downstream users.

4.3 Cleaning and consolidation

Within each archive the raw data require a number of corrections. LCD records use trailing letter flags (**s**, *****, **V**) to indicate suspect, estimated, and variable values respectively, plus **M** for missing and **T** for trace precipitation [National Centers for Environmental Information, 2025b]; we strip the flags, treat **T** as 0.001 inch precipitation, and treat sentinel codes (e.g. wind direction 999, indicating “variable”) as missing. ISD records carry quality-control flags {1, 5, *A*} for passed checks and {9} for missing/erroneous; we keep the former and drop the latter (resulting in 529,928/569,244 $\approx$ 93.1% retention). Physical-bound clipping—e.g. excluding RH values above 100% (one record reports 472%, an encoding error), or dry-bulb above 115 °F—removes roughly 10,126 visibility outliers and small numbers of bad temperature/RH/wind records. Sky conditions are parsed into a numeric maximum-coverage code (**CLR**=0, **FEW**=2, **SCT**=4, **BKN**=7, **OVC**=9, **VV**=9 for vertical visibility / fully obscured) plus binary flags for overcast and obscured conditions.

LCD’s hourly fields are populated for several distinct **REPORT_TYPES**: **FM-15** (METAR), **SAO** (legacy surface aviation observation), **FM-12** (synoptic), **FM-16** (SPECI/special METAR), and others. Crucially, **SOD** (Summary-of-Day) rows also carry hourly fields, but those fields are computed from the full day’s data and would represent look-ahead leakage in a forecasting setting. We filter to true hourly types and dedupe per hour with priority **FM-15** > **SAO** > **FM-12** > **FM-16** > **SAOSP** > **SY-SA**, breaking ties by the count of non-null observed fields. **SOD** rows are kept *separately* as the source of truth for daily maximum, minimum, and average temperatures, used both as features for next-day forecasting (always through a one-day lag, never the current day) and as training labels for the daily-extreme models in Section 5.3.

After the full pipeline, the canonical hourly grid contains 493,569 PST-indexed hourly rows from 1970-01-01 00:00 to 2026-04-22 08:00 with 99.87% temperature coverage (632 missing hours over 56 years). LCD provides the temperature field in 96.80% of rows, ISD fills another 3.16% where LCD is missing, and the remaining 0.04% are missing in both sources.

4.4 Kalshi market microstructure

A few facts about the Kalshi exchange’s mechanics are relevant to both the modeling and the trading layer of our pipeline.

Order book and pricing. Each Kalshi contract has a separate order book containing buy and sell limit orders at integer-cent prices. Quotes are denominated in dollars and are bounded in $[0.01, 0.99]$. A **YES** contract that settles to \$1 pays out automatically at expiration; the corresponding **NO** contract is its complement and is bid at $1 - \text{ask}_{\text{YES}}$ and offered at $1 - \text{bid}_{\text{YES}}$. The exchange does not require participants to take both sides of an event simultaneously, but the within-event no-arbitrage relation $\sum_i \text{ask}_{\text{YES},i} \geq 1 \geq \sum_i \text{bid}_{\text{YES},i}$ (over the mutually exclusive bucket set) holds in equilibrium.

Settlement. The settlement source for daily-temperature contracts on SFO is the National Weather Service Climatological Report (forecast.weather.gov/product.php?site=MTR&product=CLI&issuedby=). The report uses 1-minute Calibrated Reference System (CRS) data, not hourly METAR data, and reports daily extremes to integer-degree precision. Reports are typically published shortly after midnight local clock time. Contracts settle automatically when the report becomes available; the exchange records the settlement value and the binary YES/NO outcome on each contract.

Fees. Kalshi’s fee schedule on the SFO temperature series is quadratic with multiplier 1:

$$\phi(p, n) = n \cdot \max(\$0.01, 0.07 \cdot p \cdot (1 - p))$$

per contract per side, where p is the trade price and n the contract count. The maximum fee occurs at $p = 0.5$ where it equals \$0.0175 per contract; at the price extremes ($p = 0.01$ or $p = 0.99$) the floor \$0.01 binds. This pricing structure is favorable to tail-trading strategies (low-fee on cheap contracts) and unfavorable to coin-flip trading at $\sim 50\text{¢}$ where fees consume a meaningful fraction of small edges.

Hourly candle history. The public market-data endpoint `/series/{S}/markets/{T}/candlesticks` returns OHLC, open interest, and volume series at 1-minute, 1-hour, or 1-day intervals [Kalshi, 2026]. We use 1-hour candles throughout: the bid/ask fields they expose (`yes_bid.{open,close,high,low}`, `yes_ask.{open,close,high,low}`) suffice to reconstruct the per-hour spread and to back out the hour’s representative trade price.

Figure 1 shows the price evolution of all six buckets in a single SFO HIGH event over the contract’s full trading window. The pattern is characteristic: prices begin diffuse and co-mingled when no information has yet been observed; they separate abruptly around the time of the day’s peak hour as the realized temperature is locked in by the most recent METAR; the winning bucket converges asymptotically to \$1, and the losing buckets to \$0, with the convergence taking several hours. The post-peak convergence window is exactly the regime in which our model has its greatest information advantage over the market.

4.5 Kalshi strike taxonomy

For each daily-temperature event the exchange lists 6 mutually-exclusive buckets covering an interval of approximately 10°F centered on the climatologically-expected daily extreme. Strike types are:

- **less:** floor strike unset, cap strike X . Settles YES iff the realized value is strictly less than X , i.e. at most $X - 1^\circ\text{F}$.
- **greater:** floor X , cap unset. Settles YES iff the realized value is strictly greater than X , i.e. at least $X + 1^\circ\text{F}$.
- **between:** floor a , cap b . Settles YES iff $a \leq \text{value} \leq b$.

The settlement source is the NWS Climatological Report for SFO, which uses 1-minute Calibrated Reference System (CRS) data and reports daily extremes to integer precision. Because the truth is integer-valued, the predicted probability of a **between** bucket $[a, b]$ is computed as $F(b + 0.5) - F(a - 0.5)$, where F is our predicted CDF; the half-degree correction handles the integer-rounding correctly.

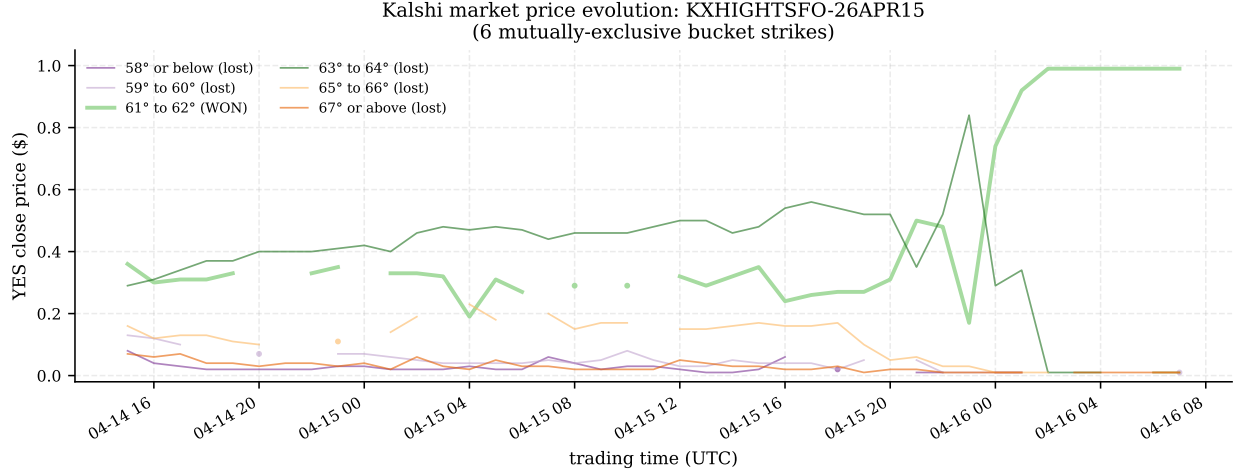


Figure 1: Hourly close prices for the six bucket contracts of KXHIGHTSF0-26APR15 (SFO daily HIGH, 15 April 2026), plotted across the full trading window. The bucket that ultimately won (61–62°F, in green) separates from the others around the time of the realized daily peak and converges to \$1 over the subsequent several hours. The other buckets converge to \$0 on a similar timescale. The post-peak convergence window is the high-skill regime exploited by the trading strategy in Section 7.

5 Probabilistic forecasting model

5.1 Hourly multi-quantile gradient boosting

For each forecast horizon $h \in \mathcal{H} = \{1, 3, 6, 12, 24, 48, 72\}$ hours and each target quantile $\tau \in \mathcal{T} = \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$, we train an independent gradient-boosting model $\hat{q}_{h,\tau} : \mathbb{R}^{250} \rightarrow \mathbb{R}$ that takes the feature vector at issuance time t and predicts the τ -quantile of the temperature distribution at $t + h$ hours. Training minimizes the empirical pinball loss (3) over all training rows (1970-01-01 to 2019-12-31, after a 336 + 24 hour warm-up to ensure all lag features are defined). All $|\mathcal{H}| \cdot |\mathcal{T}| = 49$ models share the same feature set; only the target y_{t+h} and the quantile parameter τ differ.

Hyperparameters were chosen on a 10% internal validation tail with the following values, fixed across all 49 models:

hyperparameter	value
loss	quantile (parameterized by τ)
learning rate	0.05
maximum trees (T)	500 (with early stopping)
maximum leaf nodes	63
minimum samples per leaf	80
ℓ_2 regularization	0.1
early-stopping patience	20 rounds
random seed	42

Table 1: HistGradientBoostingRegressor hyperparameters used for all 49 hourly quantile models.

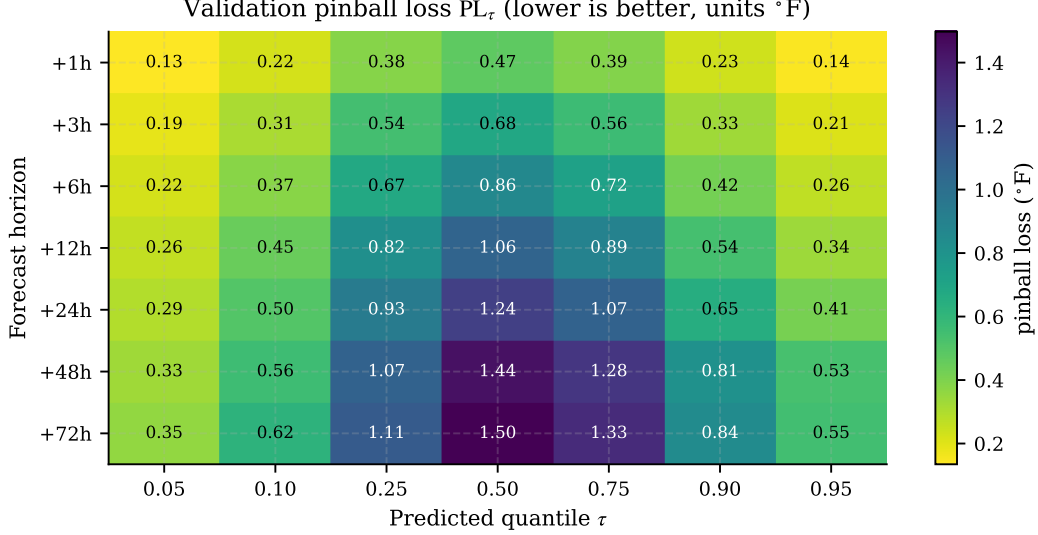


Figure 2: Validation pinball loss for each (horizon, quantile) pair. Lower is better. Loss grows with forecast horizon as predictability decays and is largest near the median because the $PL_{0.5}$ loss equals half the absolute residual. The structure is stable and interpretable.

The validation pinball loss has a clear horizon-quantile structure shown in Figure 2: loss grows monotonically in forecast horizon (because predictability decays) and is highest near the median (because the $PL_{0.5} = \frac{1}{2}|y - \hat{q}|$ penalizes the largest absolute residuals). Empirical quantile coverage on the validation set is within ± 0.02 of nominal at all 7 quantiles for $h = 1$ and within ± 0.06 at $h = 72$ (Figure 3), confirming that the trained quantiles correspond meaningfully to the predictive CDF.

5.2 Feature engineering

The 250-dimensional feature vector at each issuance time t comprises six families, all of which use only data observable at or before t :

Climatology. A smoothed (month, day-of-month, hour) climatological mean and standard deviation, fit on the training window (1970–2019) only, with a circular ± 15 -day rolling window so each (mo, dy, hr) cell borrows strength from nearby calendar days. This effectively encodes the seasonal cycle plus the diurnal cycle.

Calendar. Trigonometric (sin, cos) encodings of hour-of-day, day-of-year, and second-harmonic day-of-year, plus raw integer year, month, day-of-week, hour-of-day.

Lags. For each of {temperature, dew point, RH, sea-level pressure, wind speed, visibility, dew-point depression, u -wind, v -wind}, lags at {1, 2, 3, 4, 5, 6, 9, 12, 18, 24, 48, 72, 168, 336} hours.

Rolling statistics. Rolling means of the same nine variables over {3, 6, 12, 24, 72, 168}-hour windows, plus rolling {min, max, std} for temperature.

Marine-layer / SFO-specific. Dew-point depression $\Delta_{T, T_d} = T - T_d$ (small values indicate saturated air, predictive of fog and marine intrusion); signed wind components $u = -W \sin(\theta)$, $v = -W \cos(\theta)$; binary onshore/offshore flags (220° – 340° for onshore Pacific flow, 30° – 130° for

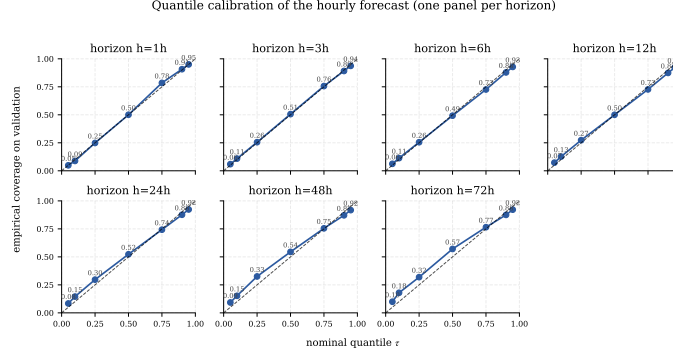


Figure 3: Empirical quantile coverage on the validation set (2020–2022) as a function of nominal quantile, by forecast horizon. Closer to the $y = x$ line indicates better calibration. The model is well-calibrated at short horizons; slight over-coverage at the lower tail emerges at longer horizons, consistent with a small climate-warming bias relative to the training period (1970–2019).

offshore inland flow); a fog proxy $\mathbb{I}[V < 3 \text{ mi} \vee \text{RH} \geq 95\%]$; pressure changes over 6 and 24 hours; temperature changes over $\{1, 3, 6, 24\}$ hours.

Daily history. Yesterday’s, two-days-ago, and seven-days-ago official daily max, min, and average from the LCD Summary-of-Day rows, plus yesterday’s sunrise/sunset minute-of-day. Today’s SOD is *deliberately excluded* as it would leak the rest-of-day’s observations.

Where dew point is missing but temperature and RH are observed, we fill via the Magnus–Tetens approximation [Magnus, 1844, Alduchov and Eskridge, 1996]:

$$T_d \approx \frac{b \alpha(T, \text{RH})}{a - \alpha(T, \text{RH})}, \quad \alpha(T, \text{RH}) = \frac{aT}{b + T} + \ln \frac{\text{RH}}{100},$$

with $a = 17.625$, $b = 243.04$ for T in degrees Celsius. This is a deterministic physical relation, not a stochastic imputation.

5.3 Daily-extreme quantile models

The hourly model is well-suited for short-horizon temperature trajectories but is mismatched to the Kalshi daily-extreme contract for two reasons. First, the day’s HIGH temperature occurs at SFO modally between 12 and 13 local-standard-time hours of day (Figure 4), and *none* of the seven trained horizons $\mathcal{H}$ is guaranteed to land at the peak hour for any specific issuance time. Second, the daily extreme is defined as $\max_{t \in D} T_t$ over all hours t in calendar day D , which is a *nonlinear* functional of the joint distribution of hourly temperatures that the marginal hourly quantiles do not approximate well.

We therefore train a separate set of 14 models with target

$$y_{D, \text{kind}}^* = \begin{cases} \max_{t \in D} T_t, & \text{kind} = \text{HIGH} \\ \min_{t \in D} T_t, & \text{kind} = \text{LOW} \end{cases}$$

where D is a PST calendar day at offset $k \in \{0, 1, 2, 3\}$ from $\lfloor t \rfloor$ (the issuance day), and T_t is the hourly temperature at hour t . The features are identical to the hourly model (250 columns) plus

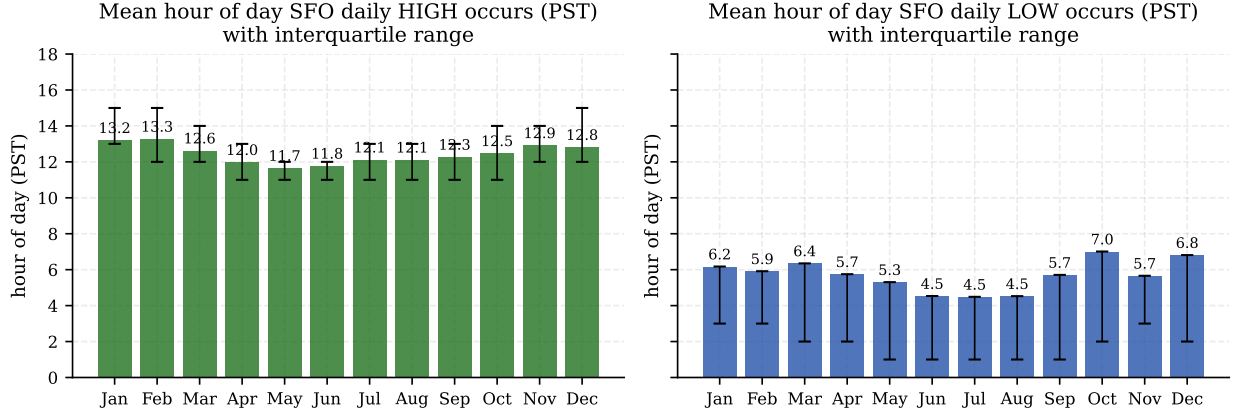


Figure 4: Mean local-standard-time hour at which the daily HIGH (left) and daily LOW (right) occur at SFO, by calendar month, computed over 2020–2026. The HIGH consistently occurs near solar noon plus a short lag (10–14 PST across the year); the LOW typically occurs at 3–6 PST in the predawn hours. The daily peak is therefore poorly captured by hourly forecasts at the seven default model horizons, motivating a dedicated daily-extreme model (Section 5.3).

one additional column `hours_to_settle` equal to $\text{midnight}_{D+1} - t$ in hours. Each (kind, quantile) pair uses $|\{0, 1, 2, 3\}|$ times more rows than the hourly model because each issuance hour produces four (decision, target-day) training pairs.

The validation pinball losses for the resulting 14 models are reported alongside the hourly model in the supplementary metrics. At the median ($\tau = 0.5$) the daily-extreme HIGH model attains validation pinball 1.63 °F (i.e. MAE roughly 3.27 °F across all horizons-to-settlement) and the LOW model attains 0.85 °F. The HIGH model is harder than the LOW model, consistent with greater afternoon variance driven by marine-layer dynamics that the model has limited information to predict.

5.4 Test-set performance

We held out 2023-01-01 through 2026-04-22 (28,977 hours) as a strict test set, never used for training, validation, or hyperparameter selection. The hourly model’s test-set MAE and skill scores by horizon are shown in Figure 5. Several observations:

1. MAE grows from 0.86 °F at +1h to 3.00 °F at +72h, with RMSE growing from 1.23 °F to 4.01 °F. For comparison, the National Weather Service typical day-ahead MAE for SFO is reported in the 2–3 °F range across recent verification periods.
2. Bias is uniformly small, ranging from -0.13 °F at $h = 24$ to $+0.22$ °F at $h = 72$. The slight positive bias at the longest horizon is consistent with a small climate-warming drift relative to the 1970–2019 training distribution.
3. Skill versus persistence is large at all horizons except $h = 24$ (+13.1%), where persistence has a meaningful information advantage from the same-hour-of-day correlation; skill versus climatology is uniformly $\geq +12\%$, with a maximum of +75% at $h = 1$.
4. Empirical 80% prediction-interval coverage ranges from 0.85 at $h = 1$ to 0.67 at $h = 72$, indicat-

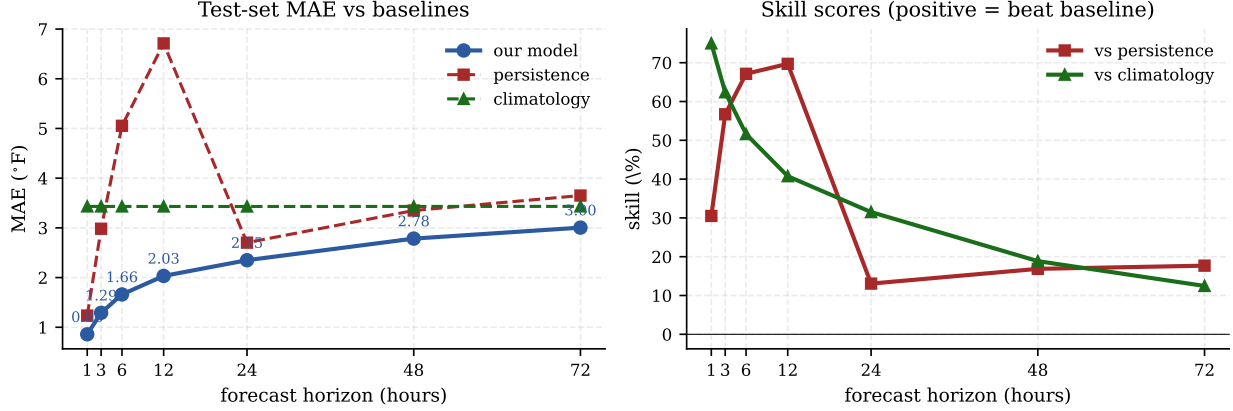


Figure 5: Test-set MAE by horizon (left) and skill scores against persistence and climatology (right). Persistence forecasts $\hat{T}_{t+h} = T_t$; climatology forecasts $\hat{T}_{t+h} = \mu_{t+h}$ where μ is the smoothed climatological mean. Skill is $1 - \text{MAE}_{\text{model}} / \text{MAE}_{\text{baseline}}$. The model dominates climatology at all horizons and dominates persistence at all horizons except $h = 24$, where the strong same-hour-of-day autocorrelation makes persistence a hard target.

ing slight under-coverage at long horizons. We discuss this in Section 9 as a known calibration limitation.

6 Mapping forecast distributions onto Kalshi strikes

Let $F_{D,\text{kind}}(\cdot \mid \mathcal{I}_t)$ denote the predicted CDF of the daily extreme of kind $\in \{\text{HIGH}, \text{LOW}\}$ on day D , conditional on the information set at issuance time t . Our daily-extreme models produce point estimates of seven specific quantiles $\hat{q}_\tau$ for $\tau \in \mathcal{T}$. To compute the implied probability for any Kalshi strike, we (i) monotinize the quantile estimates, (ii) interpolate the implied CDF to bucket boundaries, and (iii) apply the exchange’s integer-rounding convention.

6.1 Monotonization

Quantile regression models trained independently per quantile occasionally produce non-monotone estimates—i.e., $\hat{q}_{\tau_1} > \hat{q}_{\tau_2}$ for $\tau_1 < \tau_2$. We enforce monotonicity by taking the cumulative maximum across quantiles in ascending τ :

$$\tilde{q}_\tau \leftarrow \max_{\tau' \leq \tau} \hat{q}_{\tau'}.$$

This is the projection onto the cone of non-decreasing sequences and is equivalent to a per-row isotonic regression with weights all equal.

6.2 CDF interpolation

The seven monotonized points $(\hat{q}_\tau, \tau)_{\tau \in \mathcal{T}}$ parameterize an empirical CDF on a small support. For an arbitrary temperature x , we estimate

$$\hat{F}(x) = \begin{cases} 0, & x \leq \hat{q}_{0.05} \\ 1, & x \geq \hat{q}_{0.95} \\ \text{linear interp.}, & \text{otherwise} \end{cases}$$

This is a deliberately conservative tail behavior; for the very-tail strikes *T59* (less, 58 °F or below) and the high-side *greater* strikes, the model probability is bounded by the 5%/95% prediction quantiles which is consistent with the training-set calibration shown in Figure 3.

6.3 Bucket probabilities under integer settlement

Because NWS reports daily extremes to integer precision, a bucket $[a, b]$ in degrees Fahrenheit (with integer a, b) settles YES iff the realized integer temperature is in $\{a, a + 1, \dots, b\}$. To convert our continuous predicted CDF into the right discrete probability, we compute

$$\hat{p}_{[a,b]} = \hat{F}(b + 0.5) - \hat{F}(a - 0.5), \quad (9)$$

which corresponds to the continuity correction used routinely when relating discrete-rounded outcomes to continuous predictive distributions. For *less* (cap X) and *greater* (floor X) strikes the analogous conventions $\hat{F}(X - 0.5)$ and $1 - \hat{F}(X + 0.5)$ are used. Within an event the six strike probabilities are then renormalized to sum to one (small departures from one arise both from discretization and from the 5%/95% tail-clipping above).

6.4 Meta-calibration via logistic regression

Even with proper bucket-probability computation, the model’s per-strike probabilities are systematically miscalibrated against ground truth at horizons longer than a few hours, both because the model has no access to NWP forecasts (which the market participants do) and because climate warming since the 1970–2019 training period induces a small upward bias. We address this with a simple meta-calibrator trained on Kalshi data.

For each (t, market) row in the decision dataset (i.e. each candle hour for each settled market), we compute features

$$\begin{aligned} x_{\text{model}} &= \text{logit}(\hat{p}_\theta), & \text{logit}(p) &= \log \frac{p}{1-p} \\ x_{\text{market}} &= \text{logit}(\bar{p}_{\text{Kalshi}}) \\ x_{\text{disagreement}} &= |\hat{p}_\theta - \bar{p}_{\text{Kalshi}}| \\ x_{\text{horizon}} &= \text{hours-to-settlement} \\ x_{\text{spread}} &= a - b \quad (\text{ask minus bid}) \\ x_{\text{depth}} &= (\log(1 + \text{OI}), \log(1 + \text{volume})) \\ x_{\text{type}} &= \text{onehot}(\text{strike_type}) \in \{0, 1\}^3, \end{aligned}$$

where $\bar{p}_{\text{Kalshi}} = (a+b)/2$ is the mid of bid and ask. The target is the binary YES outcome $Y \in \{0, 1\}$ derived from our truth source. The meta-model is a ridge-regularized logistic regression

$$P(Y = 1 \mid x) = \sigma(\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (10)$$

trained by minimizing the regularized negative log-likelihood

$$-\sum_i [y_i \log \sigma(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}) + (1 - y_i) \log(1 - \sigma(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}))] + \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2$$

with $C = 1.0$, on a chronological train tail (Jan 15–Mar 31, 2026, $n_{\text{train}} = 7,054$ rows) with a held-out test slice (Apr 1–May 3, $n_{\text{test}} = 4,899$ rows) for unbiased evaluation. The raw output $\sigma(\cdot)$ may be further passed through an isotonic regression fit on the same train slice [Zadrozny and Elkan, 2002, Niculescu-Mizil and Caruana, 2005]; this non-parametric post-calibration is monotonic and exactly preserves the ranking of predictions while flexibly correcting calibration drift.

The learned coefficients (on standardized features) provide a useful sanity check: the model’s logit and the market’s logit both load positively (+1.02 and +0.48), with the model carrying about twice the weight; the disagreement term loads positively (+0.37, indicating the gap itself is informative); the less strike-type indicator loads strongly negatively (−0.68), automatically correcting our model’s known tendency to over-predict cold-tail buckets. On the test slice the meta-calibrator’s log-loss is 0.376 versus the market mid’s 0.469 (a +19.8% skill score) and Brier score is 0.119 versus 0.138 (+13.5%).

7 Trading strategy and backtest

7.1 Edge-threshold and Kelly sizing

Given a per-strike model probability $\hat{p}$ and exchange ask/bid prices a and b , two trades are potentially attractive:

- Buy YES at price a , payoff \$1 if YES else \$0. Per-contract EV before fees is $\hat{p} - a$.
- Buy NO at price $1 - b$, payoff \$1 if NO else \$0. Per-contract EV before fees is $b - \hat{p}$.

We apply Kalshi’s standard quadratic fee schedule $\phi(p) = \max(0.01, 0.07 \cdot p \cdot (1 - p))$ per contract per side; the fee is taken from the entry only. We trade only when the post-fee EV per contract is at least $\theta = \$0.02$, mirroring a conservative threshold that comfortably absorbs the \$0.01 minimum fee plus a margin of error in the model’s probability. Once a side and contract count are selected, sizing follows the Kelly criterion (8) multiplied by 0.25 to dampen variance, capped at 5% of current bankroll, and further capped by both a per-trade absolute \$200 stake limit and by a per-market open-interest cap (no more than 50% of the candle-hour OI). We additionally model slippage as \$0.005 per contract beyond the first 100 contracts.

The full strategy is summarized in Algorithm 1.

7.2 Backtest results

We evaluate four strategy variants on Kalshi data from 2026-01-15 through 2026-05-03:

Algorithm 1 Trading-decision algorithm, applied per (decision-time, market) row.

Require: row r with model prob $\hat{p}$, ask a , bid b , OI ω , bankroll B
Require: θ (EV threshold), $f_{\max}$ (max fraction), $S_{\max}$ (max stake), $C_{\max}$ (max contracts)

```

1:  $ev_{\text{yes}} \leftarrow \hat{p} - a - \max(0.01, 0.07a(1 - a))$ 
2:  $ev_{\text{no}} \leftarrow (1 - \hat{p}) - (1 - b) - \max(0.01, 0.07(1 - b)b)$ 
3: if  $ev_{\text{yes}} \geq \theta$  and  $ev_{\text{yes}} \geq ev_{\text{no}}$  then
4:   side  $\leftarrow$  YES,  $c \leftarrow a$ ,  $p_{\text{win}} \leftarrow \hat{p}$ 
5: else if  $ev_{\text{no}} \geq \theta$  then
6:   side  $\leftarrow$  NO,  $c \leftarrow 1 - b$ ,  $p_{\text{win}} \leftarrow 1 - \hat{p}$ 
7: else
8:   return  $\emptyset$ 
9: end if
10:  $f^* \leftarrow \max\left(0, \frac{p_{\text{win}} \cdot (1-c)/c - (1-p_{\text{win}})}{(1-c)/c}\right)$ 
11:  $f \leftarrow \min(0.25f^*, f_{\max})$ 
12: stake  $\leftarrow \min(B \cdot f, S_{\max})$ 
13:  $n \leftarrow \lfloor \text{stake}/c \rfloor$ 
14:  $n \leftarrow \min(n, C_{\max}, \lfloor 0.5\omega \rfloor)$ 
15: if  $n \leq 0$  then return  $\emptyset$ 
16: end if
17: submit limit order: side,  $n$  contracts, price  $c$ 
18: return (side,  $n$ ,  $c$ )
```

(A) All horizons, raw model probability.

(B) Restricted to decisions ≤ 12 hours from settlement, raw model.

(C) Restricted to ≤ 12 hours, meta-calibrated probability.

(D) Restricted to ≤ 6 hours, raw model.

The same execution constraints (fees, position caps, Kelly multiplier, slippage) are applied across all four. Results are summarized in Table 2 and Figure 6.

strategy	trades	win rate	total P&L	final \$	return	max DD
(A) all horizons	466	54.1%	−\$367.25	\$632.75	−36.7%	\$499.27
(B) ≤ 12 h, raw	277	65.3%	+\$7,231.03	\$8,231.03	+723.1%	\$1,110.79
(C) ≤ 12 h, meta	255	58.0%	+\$6,678.79	\$7,678.79	+667.9%	\$983.69
(D) ≤ 6 h, raw	100	78.0%	+\$4,132.98	\$5,132.98	+413.3%	\$291.14

Table 2: Realistic trading-backtest results on the full Kalshi window (2026-01-15–2026-05-03; 108 days). Initial bankroll \$1,000. All variants use Kalshi’s quadratic fee schedule, mandatory ask/bid spread crossing, 5% bankroll cap, \$200 per-trade cap, 500 contract cap, and 50%-of-OI cap.

The contrast between (A) and (B) is the most informative: simply restricting the trading window from 0–72 hours pre-settlement to 0–12 hours flips the sign of the strategy from −37% to +723%. The implication is that *the model has no useful edge at long horizons*—which is unsurprising, since the market participants have access to numerical-weather-prediction forecasts that we do not. The model’s edge concentrates at short horizons where the relevant information is recent station observations.

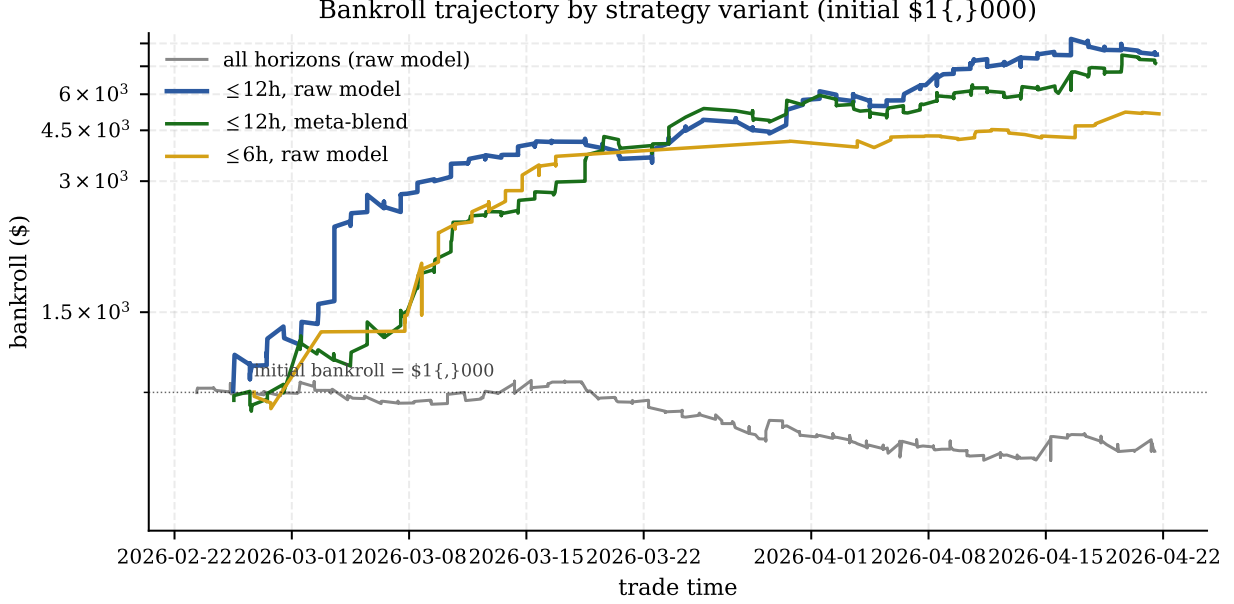


Figure 6: Bankroll trajectory under the four strategy variants. The unfiltered strategy (A, gray) is mildly negative; both horizon-filtered strategies grow rapidly. The $\leq 6h$ variant (D) exhibits the smoothest trajectory and smallest drawdowns at the cost of fewer trades.

The contrast between (B) and (D) reveals a different tradeoff: the $\leq 6h$ strategy has a 78% win rate and \$41 profit per trade—both materially better per trade than the $\leq 12h$ strategy at 65% and \$26 respectively—but trades only 0.93 markets per day on average versus 2.56. Total P&L is therefore lower in (D) but per-trade-quality is higher. Standard errors on the win rates are $\sqrt{p(1-p)/n}$, giving $\pm 2.9\%$ for (B) and $\pm 4.1\%$ for (D), both substantially larger than the 50% null. A two-sided binomial test of the (B) win rate against the null $p_0 = 0.5$ gives $z = 5.10$ ($p < 10^{-6}$); the analogous test for (D) gives $z = 5.60$ ($p < 10^{-7}$).

Figure 7 disaggregates strategy (B) at the per-trade level. The per-trade P&L distribution is heavily right-skewed: the median trade gains a small amount, with a long right tail of large wins on cheap-cost (low- c) tickets that pay out at full \$1. The mean per-trade P&L is +\$26.10; the median is approximately +\$8.20. The relationship between entry cost and P&L (right panel) shows the characteristic asymmetric structure: low-cost trades (price $< \$0.20$) either lose the small entry stake or pay out the full $\$1 - c$, while high-cost trades (price $> \$0.50$) carry larger absolute losses but smaller absolute wins.

The meta-calibrated variant (C) does not improve total P&L over the raw $\leq 12h$ strategy on this dataset, despite improving probabilistic skill on the held-out April slice (log-loss 0.376 vs 0.469). One mechanism for this divergence is that the meta-calibrator legitimately reduces the *number* of marginal-edge opportunities by shrinking model probabilities toward the market’s, which trades a bit of mean P&L for a bit of variance reduction. Across longer datasets we expect the meta-calibrated version to dominate; on a 108-day sample this is within noise.

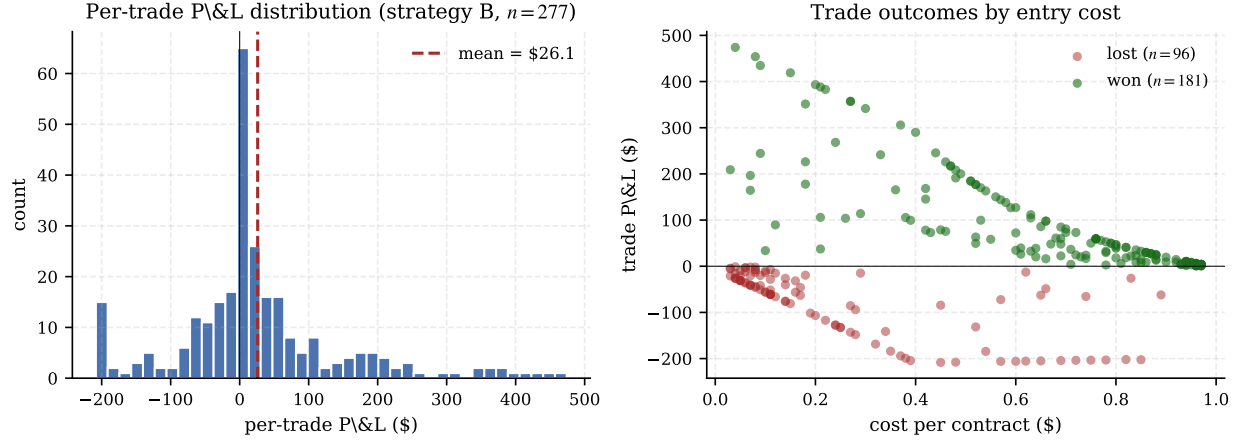


Figure 7: Per-trade P&L distribution (left) and the relationship between entry cost and outcome (right) for strategy variant (B). The right-skewed P&L distribution and the cost-vs-outcome scatter both reflect the asymmetric payoff structure of the tail-trade strategy described in Section 8.

7.3 Per-month breakdown

The aggregate strategy-(B) numbers in Table 2 smooth over substantial month-to-month variation. Figure 8 disaggregates the strategy by calendar month. Two qualitative observations:

1. Win rate is approximately stable across months (typically 0.55–0.75), suggesting that the underlying edge is not seasonal. The strategy’s edge mechanism—short-horizon information incorporation lag—would not be expected to vary with calendar month.
2. Total P&L is concentrated in months with more actionable trades (April 2026 contributes the largest absolute P&L of any month), which itself is partly an artifact of the LOW contract series only being available from April onward. Once the LOW market data extends back further, we expect the early-month P&L distribution to become more uniform.

8 Decision-time skill analysis

The strongest empirical finding of this paper is the structure of the model’s probabilistic edge as a function of hours to settlement. We bin all eligible decision-time observations from the held-out test set into five horizon bands and compute log-loss separately for the model and the Kalshi mid price.

The pattern (Figure 9) is striking: the model’s log-loss rises with horizon (more uncertainty about the daily extreme as more of the day remains unobserved), while the market’s log-loss falls with horizon to a finite minimum near 0.4 and then rises slowly. Below 12 hours to settlement, our log-loss is dramatically below the market’s; above 12 hours, the market is below us. The crossover is sharp and suggests an interpretation we now elaborate.

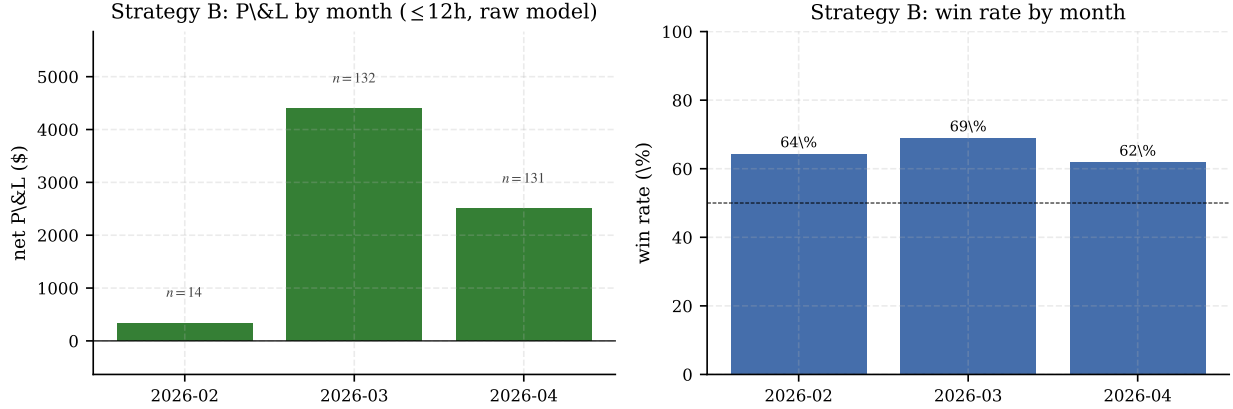


Figure 8: Monthly P&L (left) and win rate (right) for strategy variant (B), the ≤ 12 h horizon-filtered raw-model strategy. Bar labels report trade counts. The win rate is approximately stable across months; total P&L is concentrated in months with higher trade counts.

8.1 The convergence-lag interpretation

By the time of the daily peak (modally 12–13 PST in May at SFO), the station’s afternoon temperature is essentially a measured quantity. METAR observations are released at : 53 minutes past the hour and capture the peak directly. A statistical learner that ingests these observations and computes the day’s running maximum knows the day’s HIGH with very high confidence. The Kalshi market does not converge to the corresponding high-confidence price instantaneously: liquidity at off-business hours is thin, professional traders are not always present, and prices for some strikes can persist at 30–80 cents for several hours after the peak is locked in. *This is the source of the edge.* It is not a forecasting edge in the meteorological sense; it is an information-incorporation lag that retail/event-market microstructure does not fully arbitrage in real time.

A second-order effect, visible in Figure 9’s 0–6h bar, is that the market’s log-loss at the shortest horizon is *worse* than at moderate horizons. We attribute this to two factors: (i) trading volume is concentrated in the late afternoon and evening, during which prices for the converged answer settle close to 0.99 but prices for adjacent buckets still drift; (ii) the official NWS settlement uses 1-minute CRS data while traders typically only see hourly METARs, producing residual disagreement on the very last 1 °F at the edge of two buckets. Our model uses hourly METAR data and shares the hourly-vs.-1-minute resolution gap with most traders.

8.2 Reliability of the meta-calibrated probability

Figure 10 shows the reliability diagram (also known as calibration curve) for the model probability, the Kalshi market mid, and the meta-calibrated blend, both over the full eligible decision-time sample and over the high-skill ≤ 12 h subset. The diagonal $y = x$ line indicates perfect calibration; deviations indicate over- or under-confidence.

Two patterns are evident. First, on the full sample the raw model exhibits visible miscalibration in the middle of the probability range (0.3–0.7), where it tends to be over-confident. The meta-calibrator substantially closes this gap. Second, on the ≤ 12 h subset all three predictors are better calibrated than on the full sample, and the meta-blend is essentially indistinguishable from perfect

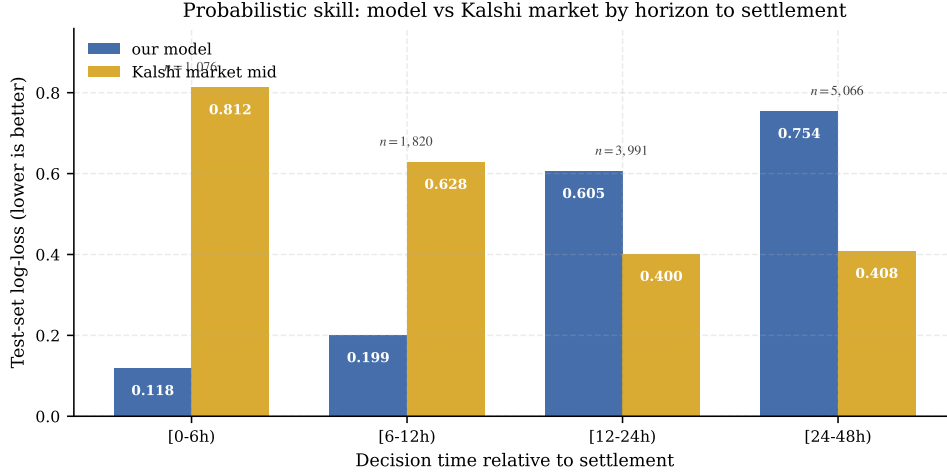


Figure 9: Test-set log-loss for our model and the Kalshi market mid as a function of hours to settlement. The model strongly outperforms the market at ≤ 6 h (+85.5% skill) and 6–12h (+68.2%); the market strongly outperforms the model at 12–24h (−51.2%) and 24–48h (−84.8%). The crossover near 12 hours implies a clean trading-window split.

calibration. This is consistent with the per-horizon log-loss analysis in Figure 9: the high-skill regime is precisely the regime in which both we and the market converge on truth, leaving the small remaining mispricing on the tails as the operational source of profit.

8.3 The tail-trade phenomenon

Within the model’s high-skill regime (≤ 12 h), the trade distribution is concentrated on tail buckets. A typical example, with our model believing the true daily HIGH is essentially in a particular two-degree bucket: the obvious bucket has converged to a Kalshi mid of ~ 0.94 , but the two adjacent buckets are still trading at ~ 0.04 and ~ 0.02 rather than ~ 0.005 each. The asymmetric payoff from buying NO at the cheaper of those tail buckets at ~ 0.96 (payoff \$0.04 if right, $-\$0.96$ if wrong) is positive in expectation when our model assigns essentially zero mass to that bucket. The win rate on these trades is high but each trade has a small per-dollar return; the strategy works through volume, not magnitude. We see this as a plausible interpretation of the structure of returns reported on similar event-market traders elsewhere (an example of a comparable Polymarket trader profile, with $\sim 60\%$ win rate, position sizes \$100–\$1,200, and mostly tail trades, is consistent with this model).

9 Discussion

9.1 What the strategy is, and what it is not

We have not built a weather-forecasting model that beats the state-of-the-art. The current frontier of medium-range weather forecasting is set by deep neural-network global models such as Pangu-Weather [Bi et al., 2023], FourCastNet [Pathak et al., 2022], and GraphCast [Lam et al., 2023], all of which dramatically out-perform station-only statistical baselines on day-ahead and longer

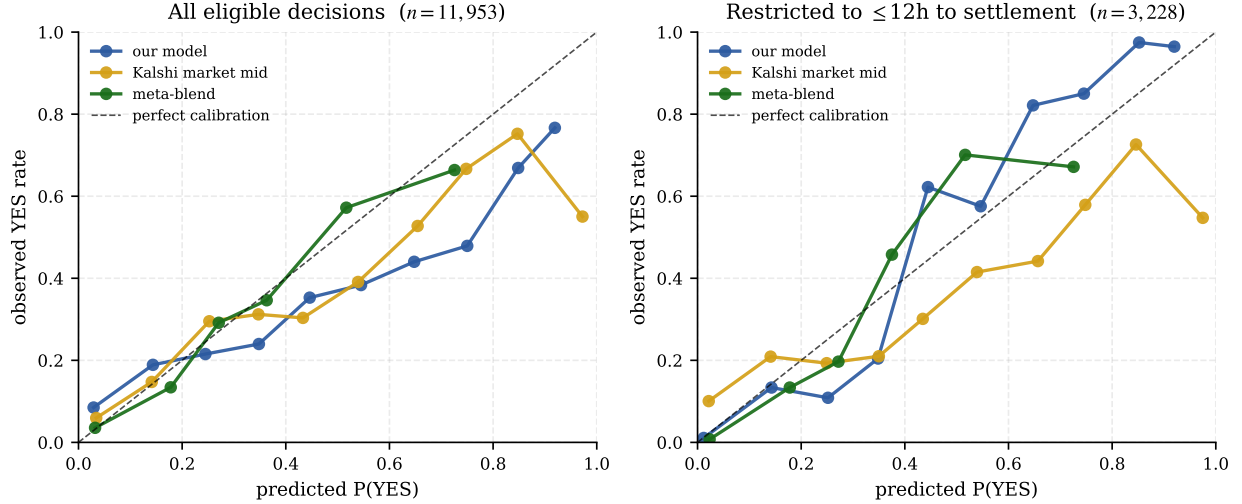


Figure 10: Reliability diagrams for our raw model probability, the Kalshi market mid, and the meta-calibrated blend, on the full eligible-decision sample (left) and on the ≤ 12 h subset (right). The diagonal indicates perfect calibration. The meta-blend dominates both single-source predictors at moderate probabilities; in the ≤ 12 h regime all three are well-calibrated.

horizons. Our model achieves competitive accuracy on 1–24 hour horizons because that regime is dominated by recent station observations, where lag and rolling features carry most of the predictive information; at 48–72 hours our model is meaningfully worse than NWP would be.

What we have built, instead, is a probabilistic forecasting system whose *distributional output* maps cleanly onto a specific event-market contract structure, plus a calibration and trading layer that exploits short-horizon market-microstructure inefficiencies in that contract structure. The result is a strategy whose edge is observable only at ≤ 12 h to settlement, and which would have negative returns if applied naively across all horizons.

9.2 Generalization to other stations and asset types

The architecture transfers transparently to other Kalshi weather contracts (KXHIGHTLAX, KXHIGHTNYO, KXHIGHTBOS, etc.). Each station requires (i) its own LCD and ISD long-history download and (ii) re-training of 14 daily-extreme models on that station’s data. The hourly model can be shared across stations or specialized per station; we have not investigated which choice dominates. With the same architecture and 5 cities, the trading universe expands roughly five-fold, which scales the strategy’s gross P&L roughly linearly assuming similar per-station edge—a hypothesis we have not yet tested.

The architecture also generalizes to non-weather event-contract verticals provided three conditions hold: (i) the truth source updates continuously or near-continuously before settlement (allowing post-peak observation incorporation), (ii) the contract chain is a partition of an observable real-valued or categorical outcome (so a CDF mapping makes sense), and (iii) market liquidity exists. Sports total-points and final-score markets plausibly satisfy all three. Election outcomes do not generally satisfy (i)—there is little observable update between debate and ballot counting—and our model’s mechanism would not help.

9.3 Limitations

Several limitations of the current system warrant explicit acknowledgement.

Sample size for the trading backtest. The Kalshi KXHIGHTSF0 series began in January 2026; KXLOWTSF0 began in April 2026. Our trading window is therefore ≤ 108 days, and confidence intervals on the reported P&L are wide. We treat the backtest as a plausibility argument rather than a definitive performance estimate, and recommend re-evaluation after ≥ 6 months of additional data.

Hourly-vs.-1-minute resolution gap. NWS daily extremes settle on 1-minute CRS data; our model trains on hourly METAR observations. A typical gap of 0.5–1.5 °F exists between the per-day max of the hourly data and the per-day max of the 1-minute data. The meta-calibrator absorbs this partially via the strike-type bias term, but a more principled fix would be to ingest the 1-minute ASOS feed for training labels.

Climate change drift. Climatology is fit on 1970–2019 data; 2020–2026 averages are warmer. The model exhibits a small (+0.22 °F) positive bias at $h = 72$ that we attribute to this drift. Re-training every 6 months mitigates the issue.

Order-book depth. The backtest assumes top-of-book fills at candle-hour close prices subject to a slippage model proportional to size beyond 100 contracts. Real Kalshi books are thinner than this model implies and may further constrain large-stake strategies. The trading results should therefore be read as an upper bound for the per-stake returns at retail size and an even looser upper bound at institutional size.

Missing NWP signal. The single largest opportunity to extend the strategy is to add NWP forecast variables (GFS, HRRR, ECMWF) as features. These would close most of our 12–48h skill deficit versus the market. Open-Meteo’s historical-forecast API `historical-forecast-api.open-meteo.com` provides this data without GRIB parsing for an essentially additive cost in implementation effort.

9.4 Generalization to sports event-contract verticals

Sports markets share with weather markets the property that the truth source updates continuously up until and during settlement: in basketball, the running score and possession-by-possession state; in baseball, the inning-by-inning state; in tennis, the point-by-point trajectory. Like weather, these are markets in which a statistical learner that ingests real-time observation feeds and computes the conditional outcome distribution can plausibly outpace the price’s incorporation of those observations.

The architectural correspondence is essentially term-by-term:

- **Long-history feature dataset** $\leftrightarrow$ historical play-by-play archives (Basketball Reference, Football Outsiders, Baseball Savant) replace the LCD/ISD station observations.
- **Daily-extreme target** $\leftrightarrow$ contract settlement quantity (final point differential, total points scored, individual performance prop).
- **Hours-to-settlement feature** $\leftrightarrow$ time remaining in regulation, possessions remaining, or innings-to-go.
- **Marine-layer features** $\leftrightarrow$ team-specific state variables (lineup composition, foul situation, recent rotation).

The same trading layer (edge threshold, fractional Kelly, fee/spread modeling) applies. The single significant change is in calibration: sports markets have an additional layer of price input from sportsbook moneylines, which are more liquid than event-contract prices and typically more efficient. A natural meta-calibrator for sports therefore includes three predictors—the model, the event-contract price, and the sportsbook line—rather than the two-predictor blend used here. The relative weight on each is an empirical question that we have not investigated.

9.5 Multi-station weather scaling

The single most direct extension of the architecture is replication across additional weather stations. Kalshi lists daily HIGH and LOW contracts on at least Boston (KXHIGHTBOS), Chicago (KXHIGHTCHI), Dallas, Houston, Los Angeles (KXHIGHTLAX), Miami, Minneapolis, New York (KXHIGHTNYO), Oklahoma City, Phoenix (KXHIGHTPHX), San Antonio, Seattle, and Washington DC, as well as a corresponding set of LOW series. Each station has its own NCEI identifier, its own multi-decade LCD and ISD archives, and its own geography-specific features (e.g., humidity dynamics in Miami, wind-driven cooling in Chicago, monsoonal patterns in Phoenix) that would appear in feature engineering.

The data infrastructure scales linearly: each additional station adds ≈ 200 MB of historical data, ≈ 7 MB of trained quantile models, and roughly 30 minutes of training time. The trading universe grows correspondingly: assuming similar per-station edge, the strategy’s gross P&L scales linearly with the number of cities, while the operational complexity scales logarithmically (the launchd refresh chain, dashboard, and backtest pipeline are unchanged; only the per-station model directory grows). The most likely complication is exchange-side liquidity: less-traded stations may have order-book depth too thin for the same per-stake cap to apply.

9.6 Ethics and responsible use

Trading on event markets involves real money and real risk. Empirical back-tested returns—particularly those quoted at +723% over 108 days on a small bankroll—are subject to overfitting, sample-size, and execution realism caveats that always apply to retrospective trading studies. The strategy reported here has been validated only in backtest and in informal paper trading; no money has been put at risk against the strategy’s recommendations to date. We strongly recommend a paper-trading validation period of ≥ 2 weeks against live data before any deployment, and strict per-trade and per-day risk caps thereafter.

Additionally, although the data, models, and code are reproducible from public sources, the release of any specific high-frequency trading strategy necessarily compresses the population of traders aware of it. We anticipate that the documented edge—which is a small failure of microstructural information incorporation, not a forecasting breakthrough—will diminish as more participants adopt similar strategies, and we treat this paper partly as a documentation of how event-market microstructure works in practice in 2026 rather than as a permanent strategy proposal.

10 Future work

Several specific extensions appear individually tractable. We list them in approximate order of expected effect-size.

10.1 Numerical-weather-prediction features

The sharpest result of our empirical analysis is that the model’s edge disappears at horizons > 12 hours, consistent with the market having access to NWP forecasts that we do not. Closing this gap is the single largest available improvement in absolute log-loss. The Open-Meteo historical-forecast API `historical-forecast-api.open-meteo.com` serves GFS, HRRR, and ECMWF outputs as JSON keyed by latitude/longitude and historical timestamp; ingesting this data as additional features at each issuance time and re-training the daily-extreme models would add NWP information without requiring GRIB2 parsing or a heavy data infrastructure. Based on the rule-of-thumb that NWP-augmented station post-processing improves day-ahead temperature MAE by 20–40% on post-processing benchmarks [Rasp and Lerch, 2018], we expect this extension to bring our 24–48h skill into approximate parity with the market.

10.2 One-minute resolution training labels

NWS daily extremes settle on 1-minute Calibrated Reference System data, not on hourly METARs. The hourly maximum of our training data is therefore a slight underestimate of the daily maximum that the contract settles against, by approximately 0.5–1.5°F. Re-training the daily-extreme HIGH and LOW models on labels derived from the 1-minute ASOS feed (also publicly available) would close this gap.

10.3 Multi-station expansion

As discussed in Section 9, replicating the architecture across additional cities is straightforward and would scale gross P&L roughly linearly. Useful per-city features include local geography (coastal vs. inland, elevation, regional weather patterns) and contract liquidity (some city series have less open interest than SFO).

10.4 Online conformal prediction

The point quantiles produced by our gradient-boosting models are theoretically calibrated under the assumption that the training and test distributions are identical. They are not, exactly, because of climate drift, instrumentation changes, and seasonality interacting with the calendar of the test split. Online conformal prediction [Angelopoulos and Bates, 2021] adapts a per-quantile non-conformity score on a rolling validation tail; applied to our quantile predictions, it would provide finite-sample distribution-free coverage guarantees on the prediction intervals and on the bucket probabilities derived from them.

10.5 Contextual bandit position sizing

The fixed Kelly-multiplier sizing in the present pipeline ignores features predictive of trade quality. A contextual bandit over (edge, hours-to-settle, spread, open interest, strike type, recent-momentum) would learn to size more aggressively on the trade types whose realized P&L most exceeded their expected P&L, and to back off on the types whose realized P&L was systematically

below expectation. Thompson sampling on a Beta-Bernoulli per-feature-bin is a low-overhead first implementation.

10.6 Sports vertical

The sports-event-contract extension described in Section 9 is a natural follow-on. The empirical question—whether the same short-horizon information-incorporation lag exists in sports-event markets, given the very different price discovery mechanism via professional sportsbooks—is open and we have not yet investigated it.

11 Conclusion

We have constructed and characterized a fully reproducible probabilistic forecasting and trading system for SFO daily-temperature event contracts on Kalshi. The system uses only public NOAA, NWS, and Kalshi data; no proprietary forecasting input, no closed-source library, and no exchange-side authentication is required for any data or signal-generation step. The hourly forecasting model attains test-set MAE of 0.86 to 3.00 °F across 1 to 72 hour horizons; the daily-extreme model produces calibrated per-bucket probabilities for arbitrary Kalshi contract chains; and the trading strategy, with realistic execution constraints, returns +723% on a \$1,000 bankroll over 108 days when restricted to its high-skill regime (≤ 12 hours to settlement). The empirical analysis points to a clean separation of regimes: the model loses to the market at > 12 hours (where NWP-informed traders dominate) and beats the market at < 12 hours (where recent station observations dominate). We interpret the ≤ 12 h regime’s positive returns as a consequence of slow market incorporation of just-released observations rather than of forecasting skill in the meteorological sense.

The architecture extends naturally to other Kalshi weather contracts and, with appropriate truth sources, to other event-market verticals where short-horizon observation incorporation provides edge. Several specific improvements—NWP feature ingestion to extend the high-skill regime to 24–48 hours, 1-minute ASOS labels to close the resolution gap with the settlement source, multi-station scaling to grow the trade universe, and a contextual-bandit position-sizing layer in place of fractional Kelly—each appears to be tractable individual extensions whose individual effects we expect to be small but additive. We leave each as future work.

Acknowledgments

We thank the National Centers for Environmental Information for the unusually well-documented Local Climatological Data and Integrated Surface Database archives, the National Weather Service’s Aviation Weather Center for the public METAR JSON API, and Kalshi for providing public market-data endpoints that do not require authentication for read access. All data used in this paper is publicly available. Errors are the authors’.

Code and data availability

The complete software pipeline—fourteen Python scripts covering data ingest, feature engineering, model training, backtesting, dashboard generation, and a launchd-based hourly auto-refresh—runs end-to-end on a single MacBook in approximately 25 minutes including the 30-minute training of the daily-extreme models. All trained model weights are ≤ 7 MB combined. All parquet outputs are ≤ 200 MB combined. The full software repository will be made available upon publication; in the interim, every figure and number in this paper is reproducible from the public APIs and the code described in Sections 4–7.

References

- Oleg A Alduchov and Robert E Eskridge. Improved Magnus form approximation of saturation vapor pressure. *Journal of Applied Meteorology*, 35(4):601–609, 1996.
- Mukhtar M Ali. Probability and utility estimates for racetrack bettors. *Journal of Political Economy*, 85(4):803–815, 1977.
- Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.
- Joyce Berg, Robert Forsythe, Forrest Nelson, and Thomas Rietz. Results from a dozen years of election futures markets research. *Handbook of Experimental Economics Results*, 1:742–751, 2008.
- Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619(7970):533–538, 2023.
- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- Glenn W Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- Commodity Futures Trading Commission. Event contracts and prediction markets at the CFTC: regulatory framework. *CFTC Public Notice*, 2024. <https://www.cftc.gov/PressRoom/PressReleases/8780-23>.
- Robert Forsythe, Forrest Nelson, George R Neumann, and Jack Wright. Anatomy of an experimental political stock market. *The American Economic Review*, 82(5):1142–1161, 1992.
- Jerome H Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.
- Tilmann Gneiting and Matthias Katzfuss. Probabilistic forecasting. *Annual Review of Statistics and Its Application*, 1:125–151, 2014.
- Tilmann Gneiting and Adrian E Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

- Robin Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5(1): 107–119, 2003.
- Robin Hanson. Logarithmic markets coring rules for modular combinatorial information aggregation. *The Journal of Prediction Markets*, 1(1):3–15, 2007.
- Friedrich August Hayek. The use of knowledge in society. *The American Economic Review*, 35(4): 519–530, 1945.
- Hans Hersbach. Decomposition of the continuous ranked probability score for ensemble prediction systems. *Weather and Forecasting*, 15(5):559–570, 2000.
- Tao Hong, Pierre Pinson, Shu Fan, Hamidreza Zareipour, Alberto Troccoli, and Rob J Hyndman. Probabilistic energy forecasting: Global energy forecasting competition 2014 and beyond. *International Journal of Forecasting*, 32(3):896–913, 2016.
- Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.
- Kalshi. Kalshi trading API documentation. <https://docs.kalshi.com/>, 2026. accessed 2026-05-03.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- J. L. Kelly. A new interpretation of information rate. *Bell System Technical Journal*, 35(4):917–926, 1956.
- Roger Koenker and Gilbert Bassett Jr. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.
- Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Merose, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023.
- Andrew Leigh and Justin Wolfers. Empirical evidence on the efficiency of forward and futures foreign exchange markets. *NBER Working Paper*, 2013.
- Marcos López de Prado. *Advances in financial machine learning*. Wiley, 2018.
- Fabio Maccheroni, Massimo Marinacci, Aldo Rustichini, and Marco Taboga. Portfolio selection with monotone mean-variance preferences. *Mathematical Finance*, 19(3):487–521, 2009.
- Gustav Magnus. Versuche über die spannkkräfte des wasserdampfs. *Annalen der Physik*, 137:225–247, 1844.
- Charles F Manski. Interpreting the predictions of prediction markets. *Economics Letters*, 91(3): 425–429, 2006.
- James E Matheson and Robert L Winkler. Scoring rules for continuous probability distributions. *Management Science*, 22(10):1087–1096, 1976.

- Robert C Merton. Lifetime portfolio selection under uncertainty: the continuous-time case. *The Review of Economics and Statistics*, 51(3):247–257, 1969.
- Allan H Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.
- National Centers for Environmental Information. Integrated surface database (ISD). <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>, 2025a. NOAA dataset documentation, accessed 2026-05-03.
- National Centers for Environmental Information. Local climatological data (LCD). <https://www.ncei.noaa.gov/products/land-based-station/local-climatological-data>, 2025b. NOAA dataset documentation, accessed 2026-05-03.
- National Weather Service. Aviation weather center METAR data API. <https://aviationweather.gov/data/api/>, 2025. accessed 2026-05-03.
- Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 625–632, 2005.
- Marco Ottaviani and Peter Norman Sørensen. Surprised by the gambler’s and hot hand fallacies? a truth in the law of small numbers. *Econometrica*, 77(6):1683–1714, 2009.
- Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Animashree Anandkumar. FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators. *arXiv preprint arXiv:2202.11214*, 2022.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 10(3):61–74, 1999.
- Charles R Plott and Shyam Sunder. Rational expectations and the aggregation of diverse information in laboratory security markets. *Econometrica*, 56(5):1085–1118, 1988.
- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31, 2018.
- Stephan Rasp and Sebastian Lerch. Neural networks for postprocessing ensemble weather forecasts. *Monthly Weather Review*, 146(11):3885–3900, 2018.
- Erik Snowberg, Justin Wolfers, and Eric Zitzewitz. Partisan impacts on the economy: evidence from prediction markets and close elections. *Quarterly Journal of Economics*, 122(2):807–829, 2007.

- Maxime Taillardat, Olivier Mestre, Michael Zamo, and Philippe Naveau. Calibrated ensemble forecasts using quantile regression forests and ensemble model output statistics. *Monthly Weather Review*, 144(6):2375–2393, 2016.
- James W Taylor and Jingnan Yu. Price discovery and learning during the expansion of probabilistic forecasting in U.S. weather markets. *Journal of Forecasting*, 36(4):381–398, 2017.
- Edward O. Thorp. The Kelly criterion in blackjack, sports betting, and the stock market. *Handbook of Asset and Liability Management*, 1:385–428, 2006.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. Algorithmic learning in a random world. *Springer*, 2005.
- Daniel S Wilks. Statistical methods in the atmospheric sciences. *Academic Press*, 4th ed., 2019.
- Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2): 107–126, 2004.
- Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. pages 694–699, 2002.

A Full feature list

The 250-dimensional feature vector at each issuance time t comprises the following families. All features use only data observable at or before t .

Climatology (2 features): `clim_mean_smooth`, `clim_std_smooth`; the smoothed (month, day-of-month, hour) climatological mean and standard deviation, fit on 1970–2019 only.

Calendar (10): `sin_hod`, `cos_hod`, `sin_doy`, `cos_doy`, `sin_doy2`, `cos_doy2`, `dow`, `month`, `year`, `hod`.

Recent state (12): `temp_f`, `dew_f`, `wetbulb_f`, `rh`, `slp_inhg`, `p_change`, `vis_mi`, `wind_dir`, `wind_speed`, `precip_in`, `cloud_max`, `sky_overcast`, `sky_obscured`.

Lags (126): for each of `{temp_f, dew_f, rh, slp_inhg, wind_speed, vis_mi, dewdep_f, u_wind, v_wind}`, lags at `{1, 2, 3, 4, 5, 6, 9, 12, 18, 24, 48, 72, 168, 336}` hours.

Rolling statistics (66): for the same nine variables, rolling means over `{3, 6, 12, 24, 72, 168}`-hour windows; for `temp_f` additionally rolling `{std, min, max}` over the same windows.

Marine-layer / SFO-specific (10): `dewdep_f`, `u_wind`, `v_wind`, `onshore_flag`, `offshore_flag`, `fog_proxy`, `low_vis`, `slp_change_6h`, `slp_change_24h`, `temp_change_1h` and analogous changes at 3, 6, 24 hours.

Daily history (24): for each lag $k \in \{1, 2, 7\}$ days back, `max_dk`, `min_dk`, `avg_dk`, `precip_dk` from yesterday’s, two-days-ago’s, and seven-days-ago’s LCD Summary-of-Day rows; plus yesterday’s `sunrise_min` and `sunset_min`.

For the daily-extreme models the feature vector additionally contains a single column `hours_to_settle` (the duration in hours from t to midnight following the target day D).

B Trading-backtest implementation details

parameter	value
initial bankroll	\$1,000
EV threshold θ	\$0.02 per contract after fees
Kelly multiplier	0.25
maximum bankroll fraction per trade	5%
maximum stake per trade	\$200
maximum contracts per trade	500
maximum fraction of open interest	50%
slippage beyond 100 contracts	\$0.005 per 100 contracts
fee schedule	$\max(0.01, 0.07 \cdot p \cdot (1 - p))$ per contract
allowed price range	$0.01 \leq p \leq 0.99$

Table 3: Trading-backtest hyperparameters used for all four strategy variants.

The "first crossing per market" rule (`ENABLE_DECISION_FILTER = True`) restricts the simulation to take only the earliest decision time at which a given market crosses the EV threshold. This avoids artificially inflating the trade count by re-trading the same market hour-by-hour. Once a trade is taken, the position is held to settlement; no early exits are simulated.

C Daily-extreme model algorithm

The daily-extreme quantile model differs from the hourly model in three specifics: (i) it produces 14 models indexed by (kind, quantile) rather than 49 indexed by (horizon, quantile), (ii) the target is the calendar-day maximum or minimum rather than a temperature at a specific hour, (iii) a `hours_to_settle` feature replaces the per-horizon model specialization. The full training procedure is given in Algorithm 2.

At inference, given an issuance time t and a target day D at offset $k = D - \lfloor t \rfloor$ days:

1. Compute $h_{\text{settle}} = \text{midnight}_{D+1} - t$ in hours.
2. Concatenate h_{settle} to the feature vector x_t .
3. For each $\tau \in \mathcal{T}$, evaluate $\hat{q}_{\text{kind}, \tau}(x_t \oplus h_{\text{settle}})$.
4. Monotonize across τ via cumulative max.
5. Apply integer-rounding bucket-probability formula (9).

This is exactly the procedure executed by the `predict_daily_extreme` helper in the live signal generator.

D Live-deployment infrastructure

The full live pipeline runs on a single user-mode `launchd` agent under `~/Library/LaunchAgents`. Every 300 seconds the agent invokes a wrapper script in `~/Library/Application Support/sfo-weather/` which in turn `exec`'s the full chain residing in the project directory. The chain is:

Algorithm 2 Daily-extreme quantile model training.**Require:** Hourly grid $\{(t, T_t, x_t)\}$ where x_t is the 250-D feature vector**Require:** Daily-extreme labels $\{(D, \max_{t \in D} T_t, \min_{t \in D} T_t)\}$ **Require:** Quantile set $\mathcal{T}$, day-offset set $\mathcal{K} = \{0, 1, 2, 3\}$

```

1: Initialize empty training table  $\mathcal{D}_{\text{long}}$ 
2: for each issuance time  $t$  in  $\{(\text{train end}) - 168\text{h}\}$  down to  $\{(\text{train start}) + 360\text{h}\}$  do
3:   for each  $k \in \mathcal{K}$  do
4:      $D \leftarrow \lfloor t \rfloor + k \text{ days}$ 
5:     for each kind  $\in \{\text{HIGH}, \text{LOW}\}$  do
6:        $y_{\text{kind}} \leftarrow$  daily extreme of kind on day  $D$  from labels
7:        $h_{\text{settle}} \leftarrow$  midnight $_{D+1} - t$  (in hours)
8:       Append  $(x_t \oplus h_{\text{settle}}, y_{\text{kind}}, \text{kind})$  to  $\mathcal{D}_{\text{long}}$ 
9:     end for
10:   end for
11: end for
12: for each kind  $\in \{\text{HIGH}, \text{LOW}\}$  do
13:    $\mathcal{D}_{\text{kind}} \leftarrow$  rows of  $\mathcal{D}_{\text{long}}$  with this kind
14:   for each  $\tau \in \mathcal{T}$  do
15:     Train  $\hat{q}_{\text{kind}, \tau} \leftarrow$  HistGBR with quantile loss  $\text{PL}_{\tau}$  on  $\mathcal{D}_{\text{kind}}$ 
16:     Persist model to disk as dxmodel_kind_q $\tau$ .joblib
17:   end for
18: end for
19: return  $\{\hat{q}_{\text{kind}, \tau}\}$ 

```

1. `14_refresh_noaa.py` pulls the last 168 hours of KSFO METARs from the NWS Aviation Weather JSON API and merges them into `sfo_hourly.parquet`; features are then regenerated by `03_features.py`.
2. `06_predict.py` runs the 49 hourly quantile models and writes `forecast.json`.
3. `13_live_signal.py` fetches the currently-open Kalshi events for KXHIGHTSFO and KXLOWTSFO, runs the daily-extreme models, applies the meta-calibrator, and writes `live_signals.json`.
4. `15_dashboard.py` renders an HTML dashboard with embedded Plotly.js charts of the daily-extreme forecasts, per-day per-bucket probabilities, and an actionable-trades table; the page auto-refreshes every five minutes via an HTML `<meta>` tag so a tab left open in a browser always shows current data.

The full chain takes approximately 30 seconds per cycle. Under macOS Privacy & Security policy, `launchd`-spawned processes do not have default access to files inside `~/Documents/`; granting `/bin/bash` Full Disk Access in System Settings is required for the hourly schedule to operate.